Class 11: Structural Bioinformatics Pt 2

Laura Lu (PID: A17844089)

AlphaFold Data Base (AFDB)

The EBI maintains the largest database of AlphaFold structure prediction models at: https://alphafold.ebi.ac.uk

From last class (before Halloween) we saw that PDB had 244,290 (Oct 2025)

The total number of protein sequences in UniProtKB is 199,579,901

> Key Point: This is a tiny fraction of sequence space that has structural coverage (0.12%)

```
244290/199579901 * 100
```

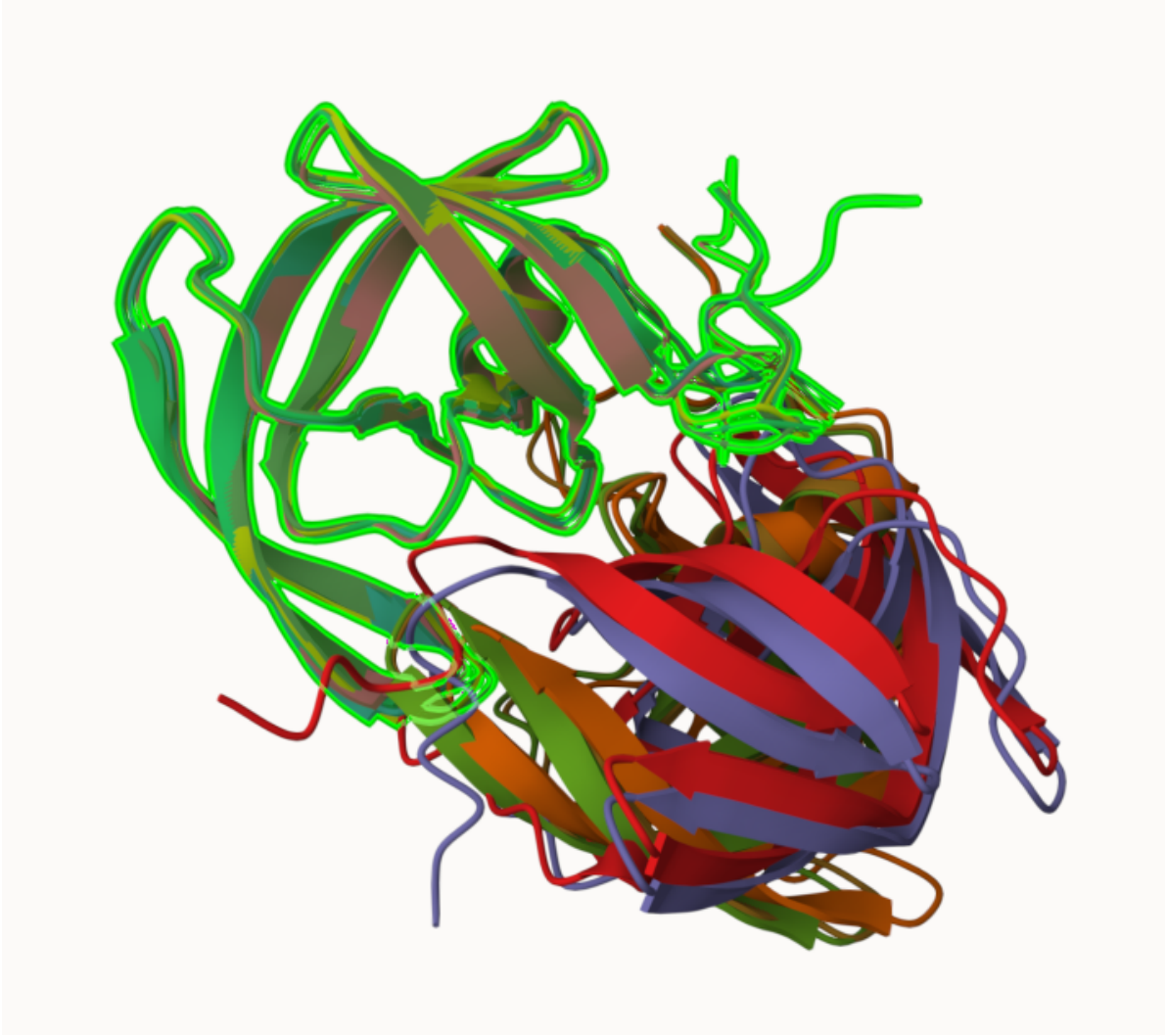
```
[1] 0.1224021
```

AFDB is attempting to address this gap...

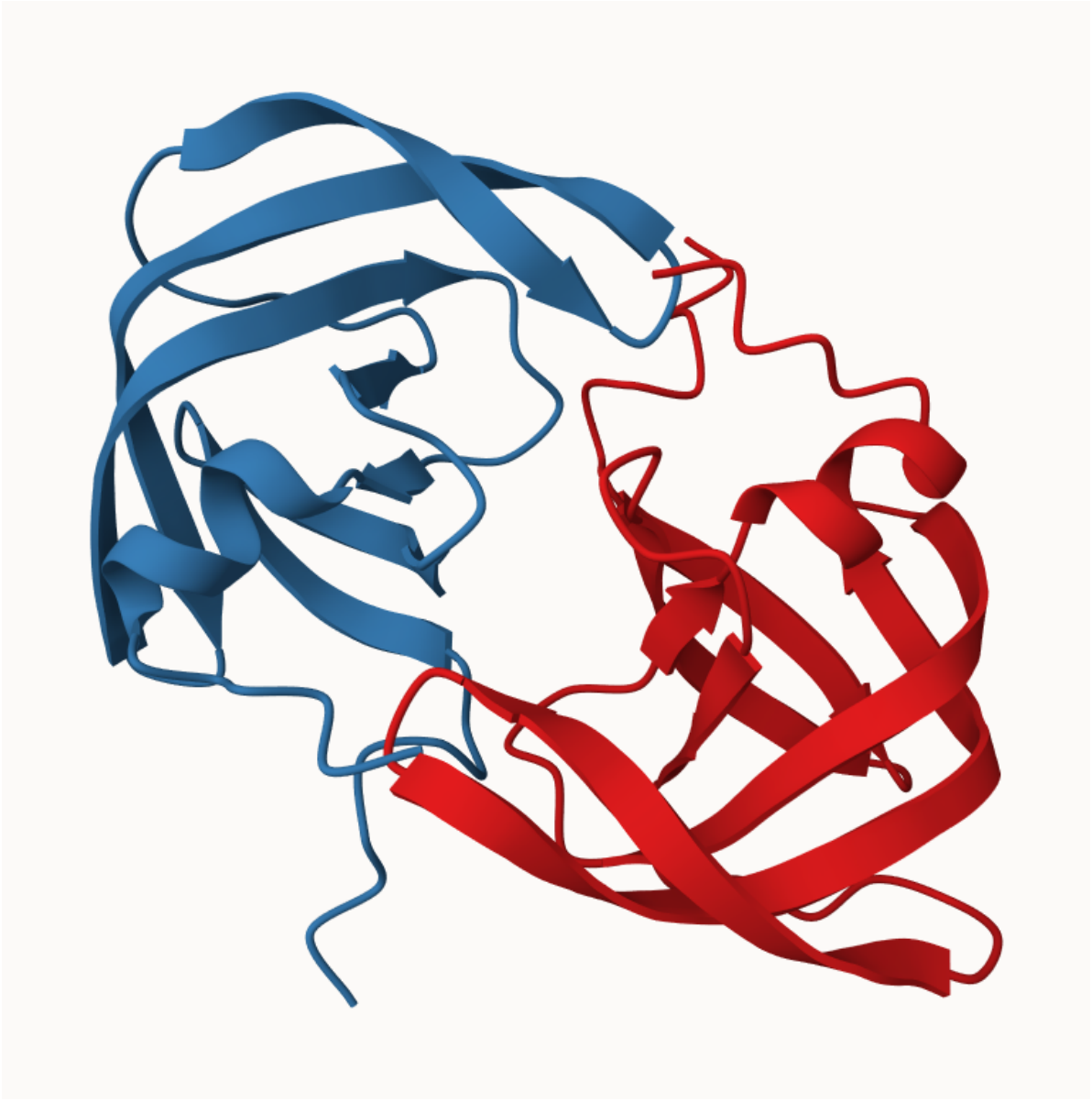
There are two "Quality Scores" from AlphaFold one for residues (i.e each amino acid) called **pLDDT** score. The other **PAE** score measures the confidence in the relative position of two residues (i.e. a score for every pair of residues).

Generating your own structure predictions

Figure of 5 generated HIV-PR models

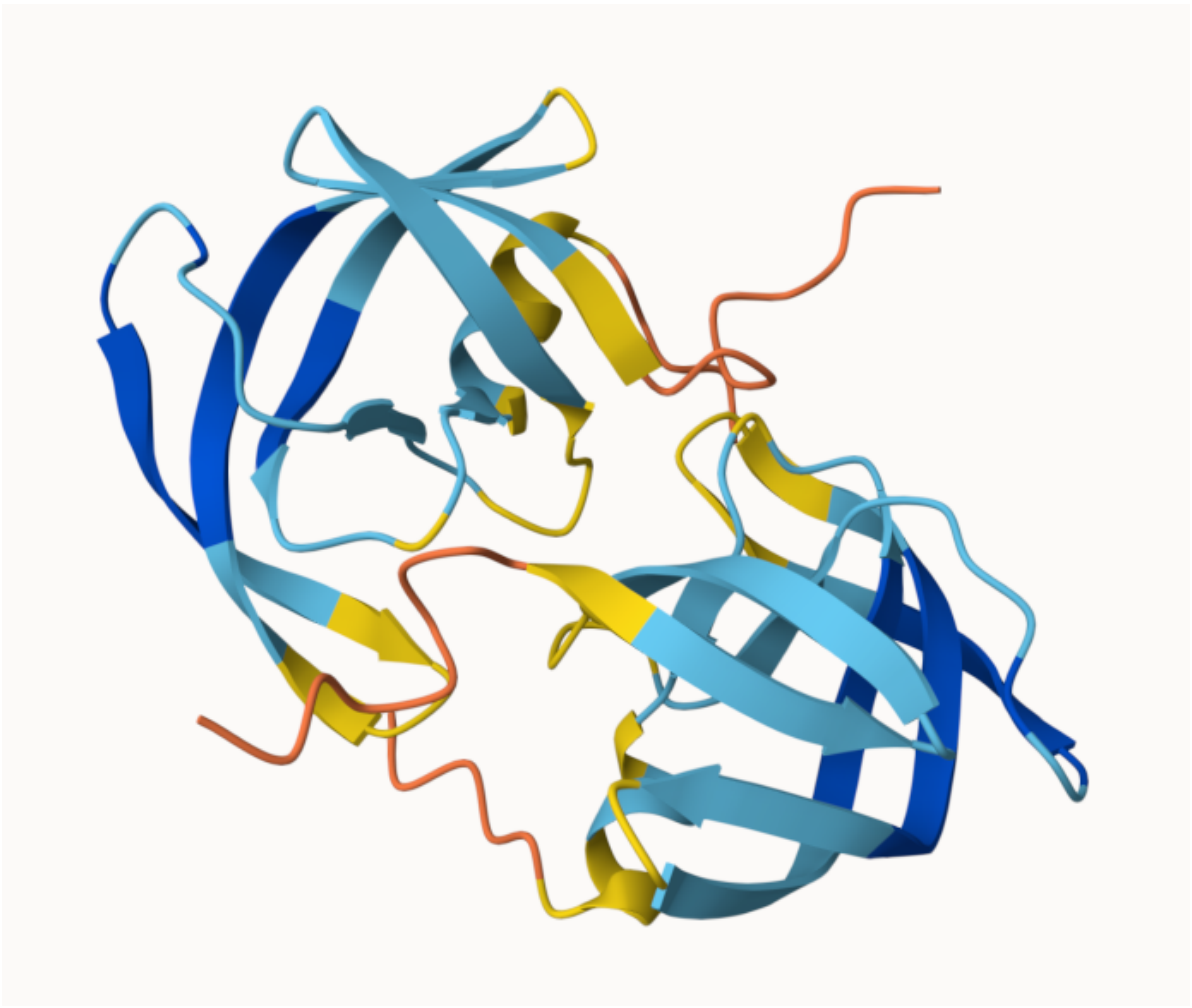


and the top ranked model colored by chain.



pLDDT score for model 1

and model 5



Custom analysis of resulting models in R

Read key result files into R. The first thing I need to know is what my results directory/folder is called (i.e. it name is different for every AlphaFold run/job)

```
results_dir <- "hivprdimer_23119:"
# File names for all PDB models
pdb_files <- list.files(path=results_dir,
                        pattern=".pdb",
                        full.names = TRUE)
```

```
# Print our PDB file names
basename(pdb_files)
```

```
[1] "HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)

m1 <- read.pdb(pdb_files[1])
m1
```

```
 Call:  read.pdb(file = pdb_files[1])

   Total Models#: 1
     Total Atoms#: 1514,  XYZs#: 4542  Chains#: 2  (values: A B)

     Protein Atoms#: 1514  (residues/Calpha atoms#: 198)
     Nucleic acid Atoms#: 0  (residues/phosphate atoms#: 0)

     Non-protein/nucleic Atoms#: 0  (residues: 0)
     Non-protein/nucleic resid values: [ none ]

   Protein sequence:
      PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
      QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
      ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
      VNIIGRNLLTQIGCTLNF

+ attr: atom, xyz, calpha, call
```

```
# Read all data from Models
#  and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_0(
```

```
hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_0
hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_0
hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_0
hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_0
.....

Extracting sequences

pdb/seq: 1   name: hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_001_alphafold2_multimer
pdb/seq: 2   name: hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_002_alphafold2_multimer
pdb/seq: 3   name: hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_003_alphafold2_multimer
pdb/seq: 4   name: hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_004_alphafold2_multimer
pdb/seq: 5   name: hivprdimer_23119:/HIVPR_dimer_23119_unrelaxed_rank_005_alphafold2_multimer
```

```r
rd <- rmsd(pdbs, fit=T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

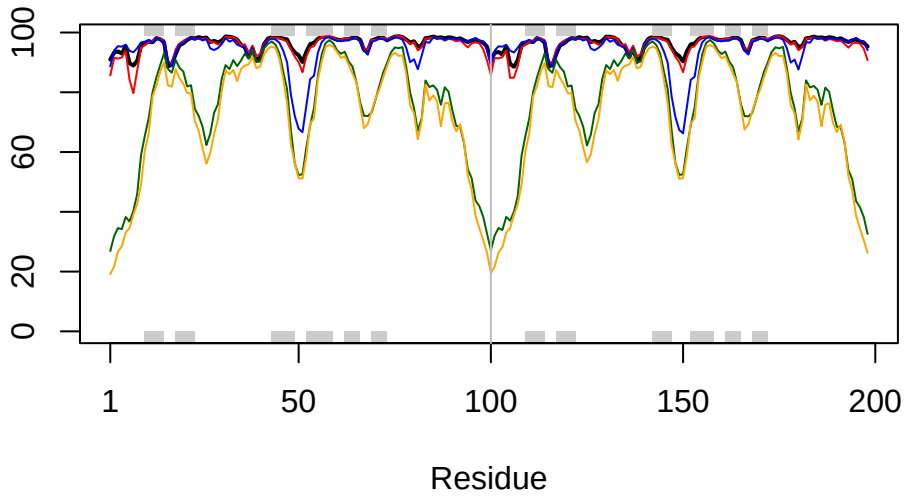
```r
range(rd)
```

```
[1]  0.000 14.929
```

```r
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```
core <- core.find(pdbs)
```

```
 core size 197 of 198  vol = 8265.928
 core size 196 of 198  vol = 1535.897
 core size 195 of 198  vol = 986.57
 core size 194 of 198  vol = 909.703
 core size 193 of 198  vol = 840.193
 core size 192 of 198  vol = 793.655
 core size 191 of 198  vol = 754.101
 core size 190 of 198  vol = 717.868
 core size 189 of 198  vol = 684.463
 core size 188 of 198  vol = 652.166
 core size 187 of 198  vol = 622.743
 core size 186 of 198  vol = 597.709
 core size 185 of 198  vol = 568.568
 core size 184 of 198  vol = 541.299
 core size 183 of 198  vol = 517.214
 core size 182 of 198  vol = 496.66
 core size 181 of 198  vol = 477.214
 core size 180 of 198  vol = 458.812
 core size 179 of 198  vol = 442.685
 core size 178 of 198  vol = 418.75
```

```
core size 177 of 198  vol = 405.786
core size 176 of 198  vol = 392.05
core size 175 of 198  vol = 378.57
core size 174 of 198  vol = 366.937
core size 173 of 198  vol = 353.608
core size 172 of 198  vol = 344.791
core size 171 of 198  vol = 332.816
core size 170 of 198  vol = 322.224
core size 169 of 198  vol = 312.69
core size 168 of 198  vol = 297.322
core size 167 of 198  vol = 282.977
core size 166 of 198  vol = 274.468
core size 165 of 198  vol = 265.976
core size 164 of 198  vol = 257.946
core size 163 of 198  vol = 249.169
core size 162 of 198  vol = 238.767
core size 161 of 198  vol = 232.897
core size 160 of 198  vol = 225.69
core size 159 of 198  vol = 220.914
core size 158 of 198  vol = 214.066
core size 157 of 198  vol = 208.998
core size 156 of 198  vol = 201.995
core size 155 of 198  vol = 197.813
core size 154 of 198  vol = 191.188
core size 153 of 198  vol = 184.362
core size 152 of 198  vol = 178.256
core size 151 of 198  vol = 173.136
core size 150 of 198  vol = 165.806
core size 149 of 198  vol = 160.026
core size 148 of 198  vol = 157.055
core size 147 of 198  vol = 150.123
core size 146 of 198  vol = 144.781
core size 145 of 198  vol = 138.674
core size 144 of 198  vol = 133.21
core size 143 of 198  vol = 127.588
core size 142 of 198  vol = 123.674
core size 141 of 198  vol = 117.858
core size 140 of 198  vol = 112.261
core size 139 of 198  vol = 108.414
core size 138 of 198  vol = 104.277
core size 137 of 198  vol = 100.881
core size 136 of 198  vol = 96.595
core size 135 of 198  vol = 93.274
```

```
core size 134 of 198  vol = 92.168
core size 133 of 198  vol = 88.258
core size 132 of 198  vol = 83.067
core size 131 of 198  vol = 79.788
core size 130 of 198  vol = 75.252
core size 129 of 198  vol = 70.613
core size 128 of 198  vol = 67.31
core size 127 of 198  vol = 64.081
core size 126 of 198  vol = 61.608
core size 125 of 198  vol = 59.449
core size 124 of 198  vol = 57.323
core size 123 of 198  vol = 54.705
core size 122 of 198  vol = 52.677
core size 121 of 198  vol = 50.588
core size 120 of 198  vol = 48.223
core size 119 of 198  vol = 45.776
core size 118 of 198  vol = 43.587
core size 117 of 198  vol = 41.969
core size 116 of 198  vol = 39.975
core size 115 of 198  vol = 37.457
core size 114 of 198  vol = 35.505
core size 113 of 198  vol = 33.764
core size 112 of 198  vol = 31.714
core size 111 of 198  vol = 29.75
core size 110 of 198  vol = 28.915
core size 109 of 198  vol = 26.991
core size 108 of 198  vol = 25.05
core size 107 of 198  vol = 23.505
core size 106 of 198  vol = 21.805
core size 105 of 198  vol = 20.109
core size 104 of 198  vol = 18.511
core size 103 of 198  vol = 16.373
core size 102 of 198  vol = 15.393
core size 101 of 198  vol = 13.322
core size 100 of 198  vol = 12.423
core size 99 of 198  vol = 10.77
core size 98 of 198  vol = 10.314
core size 97 of 198  vol = 8.903
core size 96 of 198  vol = 7.455
core size 95 of 198  vol = 5.843
core size 94 of 198  vol = 4.517
core size 93 of 198  vol = 3.186
core size 92 of 198  vol = 2.696
```

```
 core size 91 of 198  vol = 2.119
 core size 90 of 198  vol = 1.701
 core size 89 of 198  vol = 1.259
 core size 88 of 198  vol = 0.862
 core size 87 of 198  vol = 0.636
 core size 86 of 198  vol = 0.526
 core size 85 of 198  vol = 0.448
 FINISHED: Min vol ( 0.5 ) reached
```

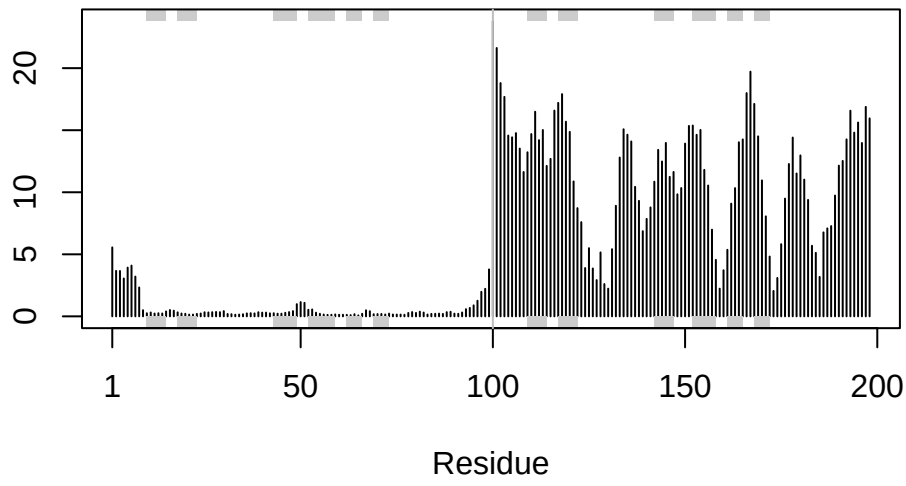
```r
core.inds <- print(core, vol=0.5)
```

```
# 86 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     8  48     41
2    50  50      1
3    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)


plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```

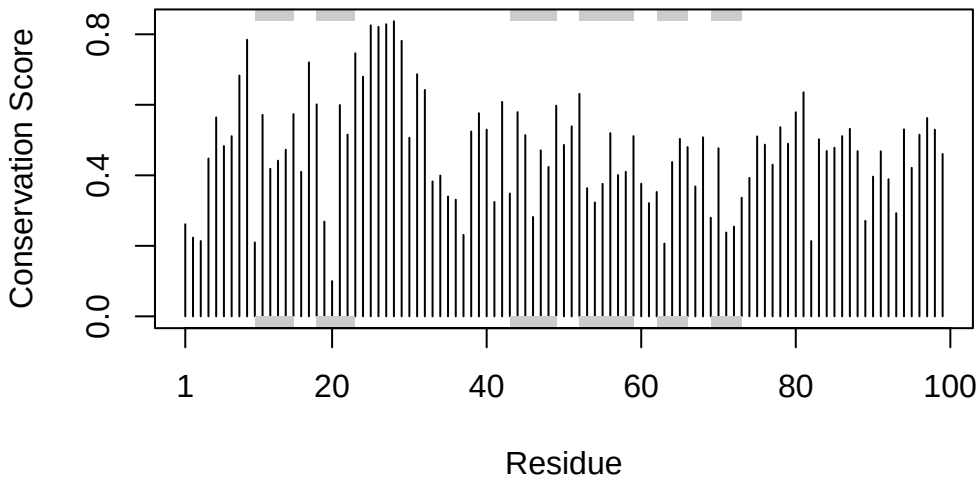



```
# Read alignment
aln_file <- list.files(path = results_dir,
                       pattern = ".a3m$",
                       full.names = TRUE)
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
sim <- conserv(aln)
```

```
plotb3(sim[1:99],
       sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```
head(m1$atom)
```

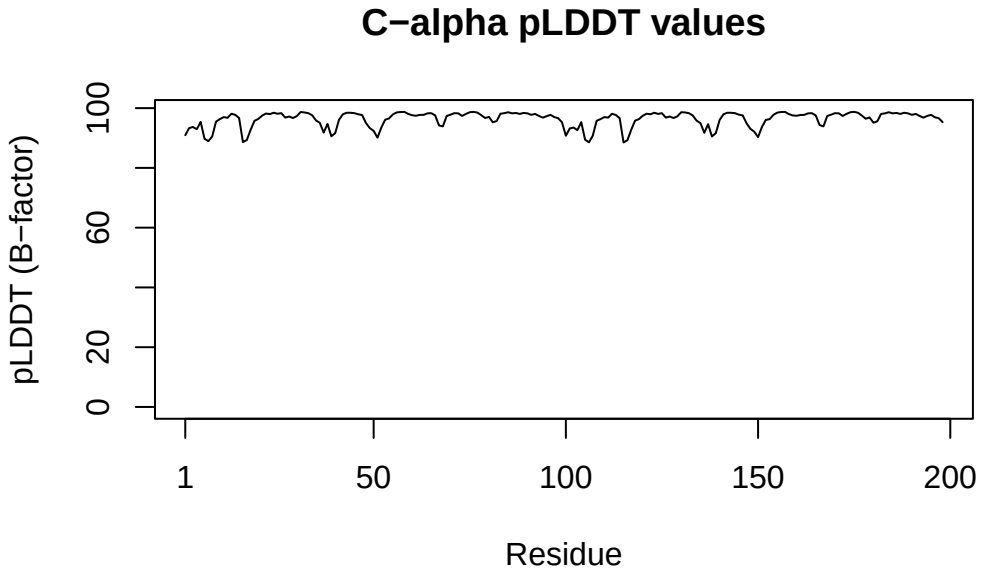
```
  type eleno elety  alt resid chain resno insert      x      y      z o     b
1 ATOM     1     N <NA>   PRO     A     1   <NA> 16.938 -3.990 -6.129 1 90.94
2 ATOM     2    CA <NA>   PRO     A     1   <NA> 16.938 -2.557 -6.430 1 90.94
3 ATOM     3     C <NA>   PRO     A     1   <NA> 16.438 -1.707 -5.266 1 90.94
4 ATOM     4    CB <NA>   PRO     A     1   <NA> 15.992 -2.449 -7.629 1 90.94
5 ATOM     5     O <NA>   PRO     A     1   <NA> 15.836 -2.232 -4.324 1 90.94
6 ATOM     6    CG <NA>   PRO     A     1   <NA> 15.070 -3.623 -7.496 1 90.94
  segid elesy charge
1  <NA>     N   <NA>
2  <NA>     C   <NA>
3  <NA>     C   <NA>
4  <NA>     C   <NA>
5  <NA>     O   <NA>
6  <NA>     C   <NA>
```

```
m1$atom$b[m1$calpha]
```

```
  [1] 90.94 93.31 93.69 92.94 95.38 89.75 88.94 90.50 95.50 96.31 97.00 96.75
 [13] 98.12 97.75 96.69 88.62 89.31 92.75 95.75 96.38 97.50 98.19 98.00 98.44
 [25] 98.12 98.31 96.81 97.19 96.69 97.31 98.69 98.56 98.31 97.62 95.88 95.06
```

```
 [37] 91.81 94.69 90.56 91.62 96.12 97.94 98.44 98.44 98.31 97.94 97.69 95.00
 [49] 93.31 92.38 90.12 93.50 96.12 96.56 97.88 98.56 98.69 98.75 98.06 97.62
 [61] 97.44 97.75 97.75 98.25 98.31 97.56 94.19 93.88 97.38 97.81 98.31 98.25
 [73] 97.38 98.06 98.62 98.75 98.50 97.62 96.69 97.06 95.25 95.69 98.12 98.31
 [85] 98.62 98.25 98.38 98.06 98.38 98.25 97.75 98.06 97.38 96.81 97.31 97.75
 [97] 97.00 96.62 95.25 90.75 93.19 93.50 92.62 95.31 89.44 88.56 90.75 95.75
[109] 96.31 97.00 96.81 98.12 97.69 96.62 88.50 89.25 92.75 95.81 96.31 97.44
[121] 98.12 97.94 98.44 98.06 98.31 96.81 97.19 96.69 97.25 98.62 98.56 98.31
[133] 97.56 95.81 95.00 91.75 94.62 90.50 91.62 96.06 97.94 98.44 98.44 98.31
[145] 97.81 97.56 94.94 93.06 92.12 90.31 93.69 96.06 96.31 97.75 98.50 98.69
[157] 98.75 98.00 97.56 97.44 97.69 97.75 98.25 98.31 97.56 94.31 93.88 97.38
[169] 97.81 98.31 98.25 97.38 98.06 98.62 98.75 98.44 97.50 96.44 96.88 95.12
[181] 95.56 98.06 98.25 98.62 98.25 98.38 98.06 98.44 98.25 97.75 98.06 97.38
[193] 96.81 97.38 97.75 96.94 96.62 95.31
```

```
plotb3(m1$atom$b[m1$calpha],
       typ = "l",
       ylab = "pLDDT (B-factor)",
       main = "C-alpha pLDDT values")
```



Residue conservation from alignmnet file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                       full.names = TRUE)
```

Read this into R

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

How many sequences are in this alignment

```
dim(aln$ali)
```

```
[1] 5397  132
```

We can score residue conservation in the alignment with the conserv() function.

```
sim <- conserv(aln)
```

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```